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Silver jubilee review

Freshwater fisheries research in New Zealand: processes, projects, and people

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Abstract The story of the discovery of New Zealand's freshwater fishes and the development of research on both the indigenous fauna and the exotic fauna that support most recreational fisheries in New Zealand is traced. It forms the background to a more substantive account of the processes that took place over the past 25 years in the development of research, the projects that were undertaken, and the key people involved in these events. This 25-year period, which equates with the life of the *New Zealand Journal of Marine and Freshwater Research*, is notable for the following fields of activity: a significant flowering of interest and knowledge of the hitherto poorly known indigenous fauna; the application of the in-stream flow incremental methodology; increased research on eels prompted by the development of a commercial wild eel fishery; major investment in research on chinook salmon populations leading to the development of salmon aquaculture research and technologies; problems with the commercial capture of salmon at sea; proliferation of major environmental impact studies that were provoked by heavy pressure for exploitation of New Zealand's rivers for irrigation and hydroelectricity; research on the use of large herbivorous Asiatic carps for biological control of excessive growth of aquatic macrophytes and plankton blooms; and concerns about the impacts of exotic fishes on the indigenous fauna.

Keywords New Zealand; history; research; freshwater fisheries; review; trout; eel; salmon; indigenous fish

INTRODUCTION

The initiation of a *New Zealand Journal of Marine and Freshwater Research (NZJMFR)* in the mid-1960s reflects an attitude of that time that fisheries were of growing importance to New Zealand, both economically and environmentally, and that a medium for publication of research results relating to these fisheries was needed. Australia had initiated a similar journal more than 15 years earlier, where a number of New Zealand studies, including some in freshwater fisheries, had been published (Burnet 1952; Cunningham et al. 1953); New Zealand's other scientific journals, such as *Transactions of the Royal Society of New Zealand* and *New Zealand Journal of Science*, were attracting increasing numbers of papers on aquatic sciences including a significant number of freshwater fisheries papers. It seems, however, that establishment of *NZJMFR* was based more on an anticipated expansion in marine and freshwater studies than to meet an already existing need; thus expansion of marine and freshwater research and establishment of the journal occurred at about the same time. To understand the place of freshwater fisheries science in the context of the above, it is necessary to delve back into colonial history.

BACKGROUND TO THE DEVELOPMENT OF FRESHWATER FISHERIES RESEARCH

Maori fisheries

The Maori were, of course, well aware of the country's indigenous freshwater fishes, and exploited them as foods very widely, particularly eels—especially when living or travelling inland, away from richer coastal marine resources (Hiroa 1921; Best 1929).

There is some debate among anthropologists about the role of eels in the traditional, pre-European Maori economy (Marshall 1987) owing to the paucity of eel skeletal remains in Maori middens. However, the high abundance of eels in our rivers, lakes, and swamps; the large variety of names applied to eels by the Maori (Best 1929); the continued interest of

modern Maori in eels as a food; and the sparseness of foods for inland living Maori make highly likely the sorts of scenario presented by Anderson (1982, 1986), in which eels were a major motivation for seasonal shifts of populations.

Other freshwater fish were also important to the Maori economy, especially: inanga (*Galaxias maculatus*), both in the upstream migrating whitebait phase, and the downstream migrating ripe adults; smelts (*Retropinna retropinna* and *Stokellia anisodon*) that enter New Zealand rivers during the spring and summer; lampreys (*Geotria australis*) that also enter rivers during spring; grayling (*Prototroctes oxyrhynchus*); and an array of smaller species.

These values ascribed by the Maori to freshwater fisheries form a small but nevertheless highly significant and culturally important facet of current Maori claims for rights promised them by the Treaty of Waitangi, signed in 1840.

Early exploration and settlement

Abel Tasman made no natural history observations in New Zealand when here in the mid-1600s, but later explorers and colonists, beginning with James Cook (whose naturalists discovered the giant kokopu, *Galaxias argenteus*) soon became aware of the native fauna, in two capacities: as food—the colonists adopting and applying more advanced materials to the capture of freshwater fish; and as indigenous fish species—of interest to ichthyologists. With regard to the latter, these “biological novelties” were at first sent back to ichthyologists mostly at the British Museum, for description (to Leonard Jenyns, Albert Günther, John Gray, John Richardson, and others); but before long the colonial community acquired the knowledge, skills, and confidence to deal with these matters themselves, and scientists like James Hector, Frederick Hutton, and Julius von Haast described new species from our waters (Hector 1871; Haast 1873, 1874; Hutton 1873). And there were others of lesser note, among whom surveyor Francis Clarke stood out (Clarke 1899). Discovery and description of the indigenous fauna continued well into the first half of the 20th century.

Meanwhile, the colonists, and in substantial measure these same three colonial leaders in science, were involved in establishing acclimatisation societies (Hector in Otago and Wellington, Hutton in Auckland, and Haast in Canterbury). They were active in introducing into New Zealand all kinds of exotic plants and animals. Freshwater fishes were among the species brought to colonial New Zealand, though they arrived later than most mammals, birds, and

plants. This is not surprising, given that they were more troublesome to transport from Europe in the small sailing ships then in use. In spite of these difficulties, however, freshwater fishes did arrive in the mid-colonial period, the first being goldfish (*Carassius auratus*) in 1864, with trouts, salmon, and diverse other fishes following in later years of the 19th century (Thomson 1922; McDowall 1990). Some soon became well-established, especially rainbow and brown trout (*Oncorhynchus mykiss* and *Salmo trutta*), and became the subject of scientific investigation. William Arthur, Otago Commissioner of Crown Lands, described the establishment of brown trout in Otago (Arthur 1879, 1882, 1884), and Hector assisted in identifying salmonids taken in New Zealand waters, as the colony struggled to know which species had succeeded (there was particular concern about salmon—Hector 1906, 1907).

Beginnings of fisheries management and science

Up to the late 19th century there was no official or organised fisheries science in New Zealand. In 1899 Government appointed Lake Ayson its first Chief Inspector of Fisheries. Ayson was no scientist, having begun his working career as a rabbit in the Waitaki Valley (Ayson n.d.), before his appointment as trout hatchery manager for the Wellington Acclimatisation Society in 1887. Evidently, Ayson learned about trout and trout culture from the English fish culturist F. S. Pillans, whom he met when growing up near Balclutha (Ayson n.d.). Ayson's appointment marked an explicit recognition of the need for management of our fisheries, and with his strong background in freshwater fisheries, Ayson naturally focused substantially on these resources. He went to North America and Europe in 1899 to investigate a variety of fisheries questions; among these were prospects for introducing new fish species to New Zealand to augment our existing fauna. Ayson (1899) reported on diverse species for introduction, including English sole and lobsters; but he also made a strong case for bringing Pacific salmon (*Oncorhynchus*) and whitefish (*Coregonus*). Clearly he was much more persuasive with these, as during the early 1900s Government made a significant investment in bringing ova of fish such as quinnat salmon (*O. tshawytscha*), Atlantic salmon (*Salmo salar*), and whitefish (*Coregonus* sp.) here. This was done with varying success (Thomson 1922; McDowall 1990). Ayson was not alone, as Thomas Donne, ambitious founding Head of the newly formed Tourist and Health Resorts Department, also arranged for importation of another salmonid, the mackinaw or lake trout (*Salvelinus*

namaycush). Thus, from the 1860s through the early 1900s, additional species were being introduced, and a few of them (primarily brown trout, rainbow trout, and quinnat salmon) became well established. They supported highly valued, primarily recreational, fisheries that attracted anglers from all parts of New Zealand as well as overseas (hence the interests of Donne and his Tourist Department).

Ayson (1912, 1913) stressed the need for fisheries management to be based on scientific knowledge and to use scientific principles. In this he was not alone. During the early 1900s, the incredibly productive and seemingly inexhaustible trout fisheries of New Zealand began to show signs of stress. It was not known then that during their establishment, salmonid fisheries typically pass through "boom" and "bust" phases, but this was precisely what was happening. The fish became smaller and were in poor condition; and anglers became dissatisfied, even though angling was still among the best anywhere in the world. But it was not as good as it had been, and anglers had become accustomed to catching large numbers of very large trout. Joining Ayson in demands for scientific research was Leonard Tripp, for many years Chairman of the New Zealand Acclimatisation Societies Association. Tripp also saw the need for science and in 1912 he suggested that: "The time had come when the Government should engage for a term of years an expert with a scientific training to come to New Zealand, study its fish, and advise the authorities what to do to keep up the standard of fishing" (Tripp 1912). He argued this repeatedly in subsequent years.

From these beginnings developed various research initiatives, most of them structured around the declining quality of New Zealand's trout fisheries and the supposed problem of depleted food stocks. Government brought to New Zealand Australian entomologist Robert Tillyard to investigate the state of trout foods (Tillyard 1920). Dominion Museum biologist William Phillipps investigated foods available for trout in lakes around Rotorua and Taupo (Phillipps 1924a). Edward Percival came to New Zealand during the late 1920s to a post as Professor of Zoology at Canterbury University College and provided much stimulus to research, especially in the Canterbury region, but also more widely (Percival 1932). And Alfred Hefford replaced Ayson as Chief Inspector of Fisheries in 1926, coming here with a degree in science. Both of these Englishmen joined Leonard Tripp in urging scientific research on fisheries, with considerable success. During the late 1920s the Wellington and North Canterbury

Acclimatisation Societies established their own research committees and provided scholarships for students to study primarily trout (Phillips 1929, 1931; Parrott 1927, 1929, 1932, 1933, 1934, 1971). Workers who later influenced the direction of fisheries biology in New Zealand, such as Derisely Hobbs and Gerald Stokell, were involved in the early Research Committee of North Canterbury. In the early 1930s, the New Zealand Acclimatisation Societies Association established a Central Research Committee (with Percival as Research Director and Hefford its Chairman), which brought together the developing threads of fisheries research from the various acclimatisation societies. This developed into a regionally funded but centrally organised research group, of which Hobbs became a central figure. Hobbs travelled widely through New Zealand, advising many acclimatisation societies on factors affecting their trout fisheries; and he published three now internationally famous studies of our fisheries (Hobbs 1937, 1940, 1948). Hobbs' studies eventually had a profound effect on the management strategies of the acclimatisation societies: they gradually changed their approach from massive and incessant stockings of trout to managing their populations by habitat protection and maintenance.

Not all freshwater fisheries research, however, was located in the Central Research Committee. Some work was being done in the universities, e.g., on the New Zealand lamprey by F. G. Maskell (1929, 1930, 1931) and on whitebait by Margaret McKenzie (1933, 1935). The Marine Department employed Leonard Hayes to study factors contributing to decline in the whitebait fishery (Hefford 1931, 1932, 1933). The recently established Department of Scientific and Industrial Research also employed geologist Harold Finlay to study the age structure of quinnat salmon stocks in New Zealand (Finlay 1930, 1972). And amateur naturalist Gerald Stokell began his lifelong studies, initially relating to salmonid biology, although he soon transferred his attention to native freshwater fishes (McDowall 1975a).

About 1935–36, the strain of funding research became too much for the acclimatisation societies and in 1937 the societies asked that staff working under the supervision of the Central Research Committee be transferred to the Marine Department. This was done, and the staff involved formed the foundations of Government managed fisheries research in New Zealand; it is notable that primarily freshwater fisheries were studied initially. The societies maintained an involvement, however, as they were required to pay to Government a 10% levy

on angling licences, which was intended to fund half the costs of trout research. K. Radway Allen, who would become internationally recognised for his study of the trout population in the Horokiwi Stream (Allen 1951) came to New Zealand from the United Kingdom as a trained fisheries biologist/biometrician, joining Hobbs in the Marine Department in the late 1930s. Although committed to funding the research, the societies complained they could not influence the choice of research subjects. As a result, the Freshwater Fisheries Advisory Council was established in 1946 to provide liaison between the Societies and Government, as well as to act as an advisory body to the Minister of Marine. This Council continued to advise Government until the late 1980s. The acclimatisation societies had majority voting strength on the Council, but Government was not obliged to take any notice of its decisions or recommendations.

Post-war activities in freshwater fisheries research

After the Second World War, research workers like Hobbs and Allen continued their rather low-key surveillance of our trout and salmon fisheries. Some work on eels, begun before the war by David Cairns (1941, 1942a, 1942b) was pursued afterwards by Max Burnet (1952, 1955); studies of coastal dune lakes were undertaken by Brian Cunningham and associates in relation to proposals to introduce additional game fishes (Cunningham et al. 1953); and the trout biologists, Hobbs and Allen, continued to service the research/advisory needs of the acclimatisation societies. During the late 1950s, through an agreement between the societies and the Marine Department, the Freshwater Fisheries Advisory Service was established to service society needs better. Funded by the societies, the Advisory Service was staffed by Marine Department technicians, with scientific supervision by Cunningham first, and later by a series of biologists brought to New Zealand for the purpose (David Lane, Ted Lewall, and Ron Little from the United States; Ceri Hopkins and Malcolm Flain from the United Kingdom). This was the situation during the early 1960s. Meanwhile there continued a few studies of native fishes in the universities, such as those of Mavis Davidson on brown mudfish, *Neochanna apoda* (Davidson n.d., 1951) and Hector McMillan on smelts, Retropinnidae (McMillan 1951, 1961). Hillary Jolly carried out some studies on lake populations of smelt (Jolly 1959, 1963, 1967).

GROWTH, EXPANSION AND A JOURNAL IN THE MID 1960S

The early-mid 1960s brought rapid change relative to periods before and soon afterwards (though not so by comparison with changes occurring in the late 1980s). The former "Fisheries Laboratory" was split to form Fisheries Management and Fisheries Research Divisions; K. Radway Allen resigned and left for Canada; he was eventually replaced as Research Director by Duncan Waugh (also from the United Kingdom). There was a significant influx of British and North American-trained fisheries biologists, many of them working on marine fisheries.

There developed an interest in freshwater fish biology in some of the universities, with MSc degree theses by Vivienne Benzie (galaxiids) and John Cranfield (bullies) at Canterbury University (Benzie 1961, 1968a, 1968b, 1968c, 1968d; Cranfield 1962) and myself (bullies) at Victoria University of Wellington (McDowall 1962, 1963, 1964a, 1965a, 1965b). At about this time Geoff Fish came from England to study problems of deteriorating water quality and excessive weed growth in the lakes of the central North Island (Fish 1963a, 1963b, 1966, 1975), and Cedric Woods was brought to New Zealand (from Ireland) to repeat Allen's (1951) study of the Horokiwi Stream, but this was never done; he became more interested in native fishes and pursued research on them (Woods 1963, 1966, 1967, 1968a, 1968b). I was appointed to the Marine Department to study the biology of whitebait (McDowall 1964b, 1965c, 1968, 1984; McDowall & Eldon 1980), replacing Roger Watson who drowned during whitebait studies in South Westland. So, at about this time there was an infusion of new biologists committed to work in freshwater fisheries, and in a real sense, this marks the beginnings of expanded research here—more or less coincident with the foundations of *NZJMR*. Progress was made on several fronts, some of them quite new, and thus significant.

TROUT STUDIES

The Freshwater Fisheries Advisory Service, under supervision by Cunningham, Lane, Hopkins, Lewall, Little, and Flain, respectively, continued to service the needs of the acclimatisation societies (as managers of our trout and salmon fisheries); the work of the Service persisted through the mid-late 1960s, though it was not finally disbanded until the early 1970s. Numerous mimeographed reports were produced (see McDowall 1982). As mentioned above, proposals to

repeat the Horokiwi study of Allen (1951) failed to eventuate; despite the high importance of recreational trout angling to the country, trout research was not such a dominating feature of freshwater fisheries work during the 1960s and 1970s. Burnet pursued studies of the inter-relationships of eels and trout (see below), Fish studied trout populations in the Rotorua lakes (Fish 1963b, 1968), and Hopkins began studies on the place of brown trout in the fish communities of some small headwater streams (Hopkins 1965, 1970, 1971). Essentially, the work of Hobbs (1937, 1940, 1948) and Allen (1951) formed the basis for trout fisheries management through the 1950s to 1970s, and relatively little trout research was done. Burnet collaborated with Edward Percival in a significant study of rainbow trout in Lake Lyndon, in the *Canterbury high country* (Percival & Burnet 1963). Much more recently, John Hayes (1984, 1987, 1988a, 1988b) elucidated problems of juvenile brown and rainbow trout recruitment into Lake Alexandrina—university work that was funded directly by the acclimatisation societies.

In recent years the “In-stream Flow Incremental Methodology” (IFIM) has begun to be applied in New Zealand to estimating habitat available to salmonids, thereby enabling the prediction of the impacts of changing flows in rivers on trout populations. Shirvell & Dungey (1983) applied these methods to brown trout, but the studies of Ian Jowett and colleagues have brought to freshwater fisheries the application of the methods in a broad and general way (Jowett 1990; Jowett & Duncan 1990; Jowett & Richardson 1990). In particular, a special issue of *NZJMR* (Vol. 24, No. 3) was devoted to the “100 Rivers Project”, of which the work of Jowett was a fundamental part. It sought to integrate IFIM approaches to river and trout population management into a broad and interdisciplinary study of rivers and river ecology in New Zealand. The use of the IFIM approach to rivers and salmonid management remains controversial, with some studies carried out here criticising its use and questioning its validity (Scott & Shirvell 1987). However, this work has now extended to the development of models shown to be capable of predicting trout populations in rivers; and these are likely to provide tools for managers to advocate rivers protection (Jowett in press). The international impacts on trout fisheries research and management may well prove as significant as the impacts of the earlier studies by Hobbs and Allen.

Another recent trout study of some note relates to the famed trout fishery of Lake Taupo. For generations this fishery had been managed entirely on an ad hoc

basis, with no understanding of the dynamics of the fishery or of its sustainable yield. The 1980s saw growing concern about exploitation rates and decline in stocks, as well as alarmist claims of excessive poaching of trout from the lake. Martin Cryer (1991) undertook an ambitious study to attempt to determine the trout production of the lake to form a basis for its management. This study was almost as notable for the extraordinarily meagre resources provided for it, as it was for the seemingly credible account of the lake's trout production that resulted. It was on the basis of this report that angler bag limits in the Taupo district were severely curtailed in recent angling seasons (reduced from 8 to 3 fish per day).

EEL RESEARCH

Burnet continued his studies of eel populations and biology, gradually moving from studying the biology of eels themselves, to studies of the interactions of eels and trout. There was a wide perception among trout anglers that the vast eel stocks of our rivers were destroying the trout population, and the acclimatisation societies invested heavily in eel eradication. The question of eel/trout relationships had actually been the primary justification for the pre-war eel studies by Cairns (1941, 1942a, 1942b), which included an experimental programme of eel eradication in Southland rivers. Burnet's work, in a small, stable, aquifer-fed stream near Christchurch, began to show that some of the ideas about eels destroying trout stocks were misconceptions, at least in some circumstances; Burnet rather shocked anglers when he showed that the presence of a healthy population of eels promoted a brown trout stock of modest numbers of large fish; by comparison a reduced population of eels allowed a large population of small fish that had little interest to anglers (Burnet 1968, 1969a, 1969b, 1969c). These ideas slowly penetrated the acclimatisation societies' strategies for “vermin” or predator management (McDowall 1990: 60).

There had been some earlier attempts to develop markets for eels, such as in the early 1900s, and then the 1930s and 1940s when acclimatisation societies were keen to see eels harvested, since this supported their goals of reducing eel stocks (McDowall 1990: 424). Some societies such as those in Otago, Wellington, and North Canterbury were involved in some of this commercialisation, and nearly all of them favoured it. However, in the 1960s and early 1970s, harvesting of eels expanded because there were good markets, and also because at about that time immigrants with family ties to eel fisheries in

the Netherlands came to New Zealand—bringing with them expertise in net making, eel fishing, and processing. But about this time Burnet was showing that perhaps reduction of eel numbers may not be as favourable to trout fisheries production as had long been assumed. Growth of eel fisheries was, in considerable part, the stimulus for further studies of eels—primarily at Victoria University where Donald Jellyman (1974, 1977a, 1977b, 1979) and Peter Todd (1974, 1980, 1981a, 1981b) both completed PhD theses on eel biology (Jellyman on juvenile ecology and Todd on adult eels). The Marine Department then employed both as eel biologists to continue this work, and to provide information to facilitate management of eel fisheries, many of which were soon under stress from heavy commercial exploitation. Jellyman worked on an unexploited eel stock in Lake Pounui, Wairarapa (Jellyman & Ryan 1983; Jellyman 1989), Todd on the exploited stocks in Lake Ellesmere, leading to the setting of catch quotas for the Ellesmere fishery (Todd 1979, 1982). Under continuing heavy exploitation from commercial fishers, and with growing concern from Maori about the decline of a valued traditional food resource, there is an ongoing need for eel research to provide background information for their sustained productivity and management (Jellyman *in press*).

ELECTRIC FISHING EQUIPMENT

Though trained as a biologist, Max Burnet had an amateur interest in electronics and turned this interest, with great skill, to the development of electrical equipment for stunning and thus catching fish (Burnet 1952). This sort of gear was crucial to the success of his eel research studies but also formed the foundation for New Zealand to develop electric fishing equipment of a standard equal to that used anywhere, and providing safety features that are unique (Burnet 1959, 1967). Our electric fishing machines, both larger generator-powered equipment as well as smaller, back-portable gear, still reflect the substantial and innovative developments that Burnet brought to the technology through the 1960s and beyond.

SALMON STUDIES

During the mid 1960s, a decision was made that information useful for managing recreational salmon fisheries could be obtained from research, and a significant investment in salmon research was initiated. This resulted in the construction of the Glenariffe Salmon Trap (Field-Dodgson & Galloway

1985), which has gradually expanded to a more general salmon research station. This decision sounded the death-knell of the Technical Field Service, and Malcolm Flain, brought to New Zealand from England in 1964 to manage the South Island Freshwater Fisheries Advisory Service, became primarily involved in salmon research at Glenariffe (Flain 1982). He also did a study of Lake Coleridge and its fish populations (Flain 1986). Hopkins was then in charge of the North Island Freshwater Fisheries Advisory Service (though it never had full-time technical support staff like the South Island did), and he moved on to studies of stream ecology (Hopkins 1965, 1970, 1971) as well as various native fish species (see below). Studies on salmon began as an attempt to understand factors affecting adult returns to the Rakaia River. The circulus patterns in the scales of the fish were used to identify the life history strategies adopted by adult fish that successfully return to fresh water. Observations on the strategies of salmon in western North America, in which fish were divided into those that go immediately to sea, those that stay in freshwater for approaching 1 year, and those that have an intermediate strategy (typically going to sea at about 3 months) were confirmed for New Zealand fish: the greatest number of fish returning were those with intermediate scale circulus patterns (first described by Finlay 1930, 1972; reconfirmed by Flain 1982).

The nature of adult salmon ecology/life history being fairly well understood, salmon research moved on from there to studies of juvenile salmon ecology in freshwater with work by Michael Field-Dodgson (1987, 1988), and Ceri Hopkins and Martin Unwin (Hopkins 1981; Unwin 1981, 1986; Hopkins & Unwin 1987). In particular, work was done to identify the downstream migration strategies of juvenile salmon, to improve understanding of factors affecting their survival and return as adult fish. This led to overturning the widely held view that flow instability and high water turbidity in the mainstem channels made the large east coast South Island rivers, such as the Rakaia and Rangitata, of little value for rearing of juvenile salmon. Studies showed there can be significant rearing and growth of juvenile salmon in mainstem rivers, enough to make the difference between a good adult return and a poor one—an issue of critical importance both to salmon angling and salmon aquaculture (ocean ranching).

COMMERCIALISATION OF SALMON

A rather abrupt change in directions of Government salmon research took place during the mid-1970s, as

interest developed, almost simultaneously, in both the hatchery enhancement of wild salmon populations and the commercial farming of salmon by ocean ranching. Enhancement and ocean ranching are both based on the release of well-grown smolts into rivers and to sea in the knowledge that adult salmon, returning to fresh water to spawn, navigate back to their natal stream (homing).

Several facets of these salmon fishery development issues seem to have developed more or less concurrently. The Council of South Island Acclimatisation Societies wanted to augment the natural salmon runs for recreational angling in east coast South Island rivers. A plan was developed to construct a hatchery on a tributary of the lower Waimakariri River, and to induce a run of salmon to the stream by annual releases of large numbers of salmon smolts. This was initially operated by the Council, but before long the project was handed over to the Ministry of Agriculture and Fisheries (Hardy 1988). At about the same time the Otago Regional Development Council commissioned a report on the potential for the commercial ocean ranching of salmon in the Waitaki River (Anon. 1976), and the New Zealand Salmon Anglers Association proposed a Salmon Fishery Development Commission to develop an ocean ranching facility at the Coleridge Power Station on the upper Rakaia River. Two companies wanted to pursue the Waitaki proposal and were persuaded by Government to form a consortium to develop commercial ocean ranching on the river (the ICI/Watties Salmon Consortium); another company had visions of a salmon ocean ranching enterprise on the Waikoropu Springs, near Takaka, in Golden Bay. Meanwhile the proposal of the Salmon Anglers Association got a frosty reception from the acclimatisation societies: they saw it as an attempt to move control of salmon fisheries away from their jurisdiction—and it was. The Commission concept went nowhere and finally some of the individuals involved in the proposal began a private, commercial ocean ranching scheme at the Coleridge site. All of these schemes were poor performers. The Ministry of Agriculture and Fisheries project at Silverstream (Hardy 1988) failed because of the lack of imaginative smolt release strategies; the Coleridge site was too far up stream to provide salmon acceptable on the fish markets; and the Waitaki facility failed to generate returns for reasons that are still not understood. No one but the proponents ever thought the Takaka proposal had much biological or economic merit, and despite every effort to develop a run, no returns of significance ever occurred there either. These were

but the beginnings of numerous other, largely failed, attempts to develop ocean ranching (Todd 1987).

It quickly became clear that there could be problems generating commercially viable returns of salmon, and that research could contribute to improved design of salmon release strategies. Through the late 1970s and early 1980s, this prospect led to research at Glenariffe, involving experimental enhancement of salmon populations through the release of hatchery-reared smolts into the rivers. A major programme attempted to optimise the size and number of salmon returning to fresh water from hatchery releases, by the careful choice of the size and the time of year that smolts were released. Large smolts released during the autumn were identified as producing the best adult returns (Unwin et al. 1989), and strategies developed at Glenariffe formed the basis for commercial ocean-ranching releases.

With the rapid proliferation of ocean ranching salmon farms, and their generally poor returns of fish, there was a heavy, continuous, and growing demand for salmon ova to provide fish for releases. Enhancement strategies soon began to increase adult returns to Glenariffe. This project was for a time jeopardised by the commercial demand for ova; in practical terms the only effective and reliable source of ova was Glenariffe, though for some time during the early years, ova were also obtained from wild fish taken in Deep Stream (Rangitata River), Winding Creek (Waimakariri River), and the tailrace of the Highbank Power Station in the middle Rakaia. But Glenariffe soon became heavily involved in the supply of salmon ova to salmon farmers, providing founding and maintenance stock for their farming operations—somewhat distorting its primary goals as a salmon research facility.

Before long salmon farmers became interested in freshwater pond rearing of salmon, and then in sea-cage rearing of the fish. With these moves towards captive rearing of quinnat salmon, the salmon research group began to develop techniques for their sexual manipulation in an attempt to avoid problems relating to the early maturation of males; this was done by the production of lines of all-female stock (Hopkins & Todd 1991). At about the time that these lines of all-female salmon were developing, Government pressures developed for research groups to approach their work on a more commercial basis. The all-female salmon became commercially valuable, and the research group became locked into the maintenance of these stocks and the annual production of all-female salmon ova for sale to salmon farmers. Overall, Glenariffe became a major commercial

source of ova for salmon farming operations, for those farms that did not produce their own. Thus there was a clear shift from a *purely research operation* that supported the management responsibilities of the acclimatisation societies for recreational angling, to an *extensively commercial, revenue-producing operation* aimed both at giving research support to commercial salmon production and at generating revenue for the research group.

SALMON CAPTURE AT SEA

The capture at sea by commercial trawlers of substantial quantities of salmon, in addition to causing profound concern among anglers, also appeared to threaten the wild stocks of fish (Unwin & Todd 1988; Todd et al. 1989). The need to understand the nature of yearly fluctuations in the wild salmon runs emphasised the importance and value of the long-term series of data on the salmon runs in the Rakaia River that were represented in the Glenariffe salmon database (1965 to present: Field-Dodgson & Galloway 1985). Commercial sea catches of salmon are a matter for continuing concern and research input; in particular, it is difficult to distinguish reductions in the runs as a result of natural variations in oceanic and river conditions, from reductions attributable to capture by commercial fishing boats. A recently negotiated agreement between salmon fisheries managers, commercial fishers, and research scientists (Beattie 1991) provides hope that the problem of the impacts of sea catches on salmon stocks has been resolved; if so, resurgence of salmon populations will depend on proper management of the recreational fisheries and on natural conditions in the rivers and at sea. The very large commercial releases of smolts into the Canterbury Bight during the late 1980s are possibly connected to the recent decline in Rakaia River salmon stocks—the inshore marine environment may be overloaded. These commercial releases have now been greatly reduced and it will be interesting to see whether a recovery in salmon stocks now occurs.

ENVIRONMENTAL IMPACT STUDIES

For decades, river developments in New Zealand had been causing losses to freshwater fisheries, especially trout and salmon. Acclimatisation society officials had long recognised these losses (Anderson 1977, 1979), but seemed unable to bring about effective mitigation. A dam on the Waitaki River truncated the huge salmon run into that river in 1935, and although

a fish pass was constructed on the dam it never worked and there seems to have been little Government commitment to making it work—if the final comments of Chief Inspector of Fisheries Alfred Hefford mean anything (New Zealand Marine Department 1937): “inevitably the presence of this obstruction must bring about considerable reduction in the total salmon propagation capacity of the Waitaki River. ... All that can be said is that the construction of the Waitaki Dam has staged an experiment in salmon and trout ecology which, though unsought, will yield valuable practical information” [none of which ever seems to have been obtained]. Hefford found it “not difficult to be philosophical about the effects of the dam”. Today, few would agree with him.

Having “wrecked” the Waitaki salmon fishery, Government then constructed a dam on the Clutha River at Roxburgh, and did the same to the Clutha salmon fishery. This time there was no attempt at a fish pass, nor any reference to “an experiment in salmon and trout ecology”. The Minister of Internal Affairs at the time took the view that truncating the Clutha salmon run at Roxburgh would prevent their invasion of trout habits in the upper river (though they were already doing so), and it is ironic that the Minister’s Wildlife Service spent extensive funds and effort over the subsequent 30 years obtaining salmon ova, raising and releasing smolts into Lake Wakatipu for the anglers of the district.

The Waitaki and Clutha cases are highly dramatic versions of what was happening to rivers throughout the country, but especially to the rivers of the eastern South Island where removal of water for irrigation was threatening the viability of fish populations. Stretches of river increasingly became dewatered in summer and autumn; river mouths closed owing to the lack of flow volumes; residual rivers heated up as flows declined; and where water was abstracted fish were entrained with it and eventually stranded on farm pastures, providing nothing better than food for seagulls. The affected acclimatisation societies complained vigorously, largely without effect.

When, however, Government proposed to harness the waters of the Tongariro for a massive hydro-electricity scheme, it quickly recognised the potential to damage perhaps the most famous trout angling river in the country; and it was realised this could not be done without provoking an outcry from anglers. In addition the river’s fishery was managed not by an acclimatisation society, but the Government’s own Wildlife Service (Department of Internal Affairs). For these reasons a substantial and detailed

environmental impact evaluation of the river, and the potential effects on it of the proposed developments, was carried out. The work was allocated to the Marine Department, and Cedric Woods was assigned to the Tongariro study (Woods 1964). This represented the first significant move by fisheries research into environmental impact evaluation, and would be the only substantial study of its type for some time. Woods (1964) suggested that, with some careful attention to detail like screens on inflows of water from the Wanganui River system that would prevent eels and lampreys from entering, the proposed scheme need not have seriously damaging effects on the Tongariro trout fishery; with this reassurance, the scheme was constructed, but seems likely to have damaged the fishery more seriously than predicted.

The Tongariro study completed, there were no further studies of this sort of any substance until the late 1970s. Then, pressure was being exerted for extensive further abstraction of waters of eastern South Island rivers for irrigation, as well as on small-medium rivers all over the country for hydro-electric power generation. In both instances these pressures were exacerbated by Government policies which then provided substantial subsidies for such works—many of which would never have gone ahead had a full economic justification of the schemes been demanded. This meant that a highly artificial economic environment was set up, which made any attempts to apply cost/benefit analysis or comparisons to the works largely fruitless. As a basis for understanding the nature of the threats to rivers, at one stage up to 56 small-medium New Zealand rivers were the target of prospective hydro dams, (e.g., Hurunui, Ahuriri, Manganuioteao Rivers). Such rivers might produce some electricity, but their contribution to the national needs for power was very small indeed (Teirney 1980). Acclimatisation societies saw these proposals as serious threats to the future of wild-river angling, and said so (Anderson 1977; Scott 1983).

It was in this environment that moves were made towards the development of more skilled and better equipped research groups involved in environmental impact evaluation, to provide the country's decision-makers with a more objective and better informed foundation upon which to make water use decisions. For a decade from the late 1970s, Government freshwater fisheries research staff were strenuously engaged in assessing the fish and fisheries values of rivers all over the country, from the Motu River in Bay of Plenty to the Clutha River in Otago. This was at a time when knowledge of the ecology of freshwater fishes was far from advanced, and when ecological

rather than environmental impact studies should have been undertaken. The developmental pressure on rivers prevented this. The passage in 1981 of the so-called Wild and Scenic Rivers Amendment of the Water and Soil Act 1981 (precipitated to a significant degree by the battle over the Aramoana aluminium smelter proposal and its potential effects on power production needs) provided the acclimatisation societies with a much-needed mechanism for obtaining some long-term protection of rivers that they particularly valued for their fisheries. Working in concert, the societies and the Ministry of Agriculture and Fisheries' freshwater fisheries research staff very quickly set about assembling dossiers of information that provided the societies with background information upon which to launch application for Conservation Orders. There were many such applications (and hearings), the societies being by far the most active group involved in rivers protection; the Rakaia River, however, was the one that took precedence over others from the perspective of fisheries protection.

The Canterbury Plains suffer serious water deficit for agriculture during summer and autumn, and for decades farmers and engineers had viewed the waters of rivers like the Rakaia and Waimakariri, flowing out to sea, as largely "wasted". They had visions of exploiting this water for irrigation, to increase the agricultural productivity (and reliability) of the plains. During the late 1970s and early 1980s, the Ministry of Works and Development, in association with farmer groups, developed plans for huge and sophisticated irrigation systems for the land between the Ashburton and Rakaia Rivers and also as far north as the Waimakariri River. These proposals recommended that the Rakaia River should be the major source of water, and that it should be abstracted down from its natural mean flow of about $200 \text{ m}^3 \text{ s}^{-1}$ (cumeecs) to a minimum flow of only 50 cumeecs. The scheme became financially viable only because of the very extensive injection of Government subsidies for both construction and operation of the proposed schemes and the additional, downstream subsidies on things like fertilisers and lamb meat and wool support prices (it was never truly economically viable). Acclimatisation societies, and related anglers groups like the Salmon Anglers Association, saw these proposals as spelling the end of the Rakaia salmon fishery, with a prospect that the mouth of the river would close at extended low flows of 50 cumeecs—especially in summer and autumn, when salmon were running into the rivers. With the salmon fisheries of the Waitaki and Clutha already heavily depleted,

anglers saw the next largest and most valued river, the Rakaia, suffering a similar fate. Because of the impending conflicts between irrigation interests and anglers, the Rakaia was clearly a candidate for extensive impact-related research, and became the largest of many impact-related studies that characterised freshwater fisheries research during the 1980s (Davis et al. 1983; Eldon & Greager 1983; Sagar 1983; Unwin & Davis 1983; Glova & Duncan 1985).

Making decisions about how much removal of water from a river is acceptable to in-river users of water is always a difficult task, and the Rakaia was no exception; being a large, braided river that floods seriously it was also a difficult river both to study and for which to make recommendations on acceptable flows. As it happened, acclimatisation societies strategically sought to circumvent making compromise decisions on how much water abstraction could be accepted without unacceptable damage to fish stocks and fisheries, by applying for a National Conservation Order. This sought to protect the fisheries values and amenities of the river as a whole. Though it took a local hearing, an appeal to the Town and Country Planning Appeal Tribunal, a further case in the High Court, and yet another in the Appeal Court, the acclimatisation societies finally won and Government issued a National Conservation Order protecting the natural values of the river system in 1988 (Davis 1988). The biological foundations for the various statutory processes of applying for and being granted the conservation order were largely the research results of the Government fisheries scientists who had been studying the river, its fauna, and fisheries from the late 1970s.

Another interesting perspective on fisheries environmental impact research can be obtained from research that was done on the lower Waitaki (below the Waitaki Dam) and Clutha Rivers (below the Roxburgh Dam). The New Zealand Electricity Department (NZED) had ambitions for the development of these two rivers for hydro-electricity. With a genuine desire to ensure that the fisheries implications of their proposed developments were properly understood, but also aware that, if they weren't, any proposals for development could be delayed or even rejected, NZED sought environmental impact research on these two rivers. At a time when there were severe financial constraints and rigid staff ceilings upon Government agencies, there were no resources available in MAF to staff and fund such studies. NZED offered to provide the funding, and staff ceiling problems were circumvented by appointing staff to

work on these rivers under "contract". Of particular research significance were the so-called "demonstration channels" in the lower Waitaki—a series of braids with flow-control headworks to provide facilities for simulating what a residual river might look and behave like, should the NZED's proposals for development of the river proceed. These studies resulted in a long series of detailed and complex reports on the lower Waitaki River (e.g., Graynoth et al. 1981; Deverall 1986; Smith & Pierce 1986; James & Deverall 1987; Palmer 1987; Pierce 1987; Rutledge 1987).

Similarly, funds and staff were provided to study the lower Clutha and its fish and fisheries for NZED, with several significant reports being produced (Jellyman 1984, 1987; Harvey & Jellyman 1986; Whiting 1986; Pack & Jellyman 1988). However, more recent changes in the structure of enterprises involved in designing and building hydro dams, and managing electricity production, as well as changes in forecasts of New Zealand's electricity needs, have meant slow progress on construction of further dams on the Waitaki and Clutha Rivers.

Proposals to construct a further thermal power station on the lower Waikato River, and then the need for Electricorp (which in part succeeded NZED) to renew its water rights to operate the existing Huntly Thermal Power Station on the river, generated demands for studies of the impacts of thermal effluents on the lower Waikato. By this time, research agencies were under pressure from Government to operate in a more commercial mode, and restrictions on budgets and staff were liberalised. An extensive series of studies of the lower Waikato River and the impacts of the existing Huntly Power Station was led by MAF scientists Jacques Boubée and Charles Mitchell, resulting in a very long series of substantial reports (summarised and discussed in McDowall in press).

ANGLER SURVEYS / DIARY SCHEMES

Obtaining some measure of the extent of exploitation of trout and salmon stocks is always a preoccupation of the managers of these fisheries, and various sorts of angler evaluations have long been a feature of freshwater fisheries research. The first major nationwide angler diary scheme was for the period 1947–52, (Allen & Cunningham 1957). Further angler surveys/diary schemes were carried out by the Marine Department in 1957–58, 1962–63, and 1967–68, and the results of these surveys were analysed by Eric Graynoth during the early 1970s (see New Zealand

Marine Department *Fisheries Technical Reports* by Graynoth and colleagues, listed in McDowall 1982).

With increasing development pressure on river fisheries, as discussed above, it was seen as a priority by both researchers and managers, that a better understanding should be obtained of the values ascribed to river fisheries. As a result, an angler survey of river fisheries was carried out in the late 1970s and early 1980s (a long series of New Zealand Ministry of Agriculture and Fisheries *Fisheries Environmental Reports*, variously authored by Laurel Teirney, Jody Richardson, and Martin Unwin through the 1980s: Teirney & Richardson in press). This survey differed from earlier ones in that it had more clearly prescribed goals, and was less preoccupied than previous ones with how many fish were caught and how big they were. It was concerned more with how much use rivers were put to by anglers; what features of the rivers anglers most valued; and it attempted to identify rivers of national, regional, and local significance from the anglers' perspectives. In addition the survey analysis also attempted to identify different qualities that made rivers popular or otherwise. It did this by asking respondents to rate the importance of such valued features as ease of access, proximity to home, sense of peace and solitude, high catch rate, large fish, scenic beauty, etc. In the analysis of returns, it became possible to identify different types of angling experience that anglers were obtaining and valuing in rivers.

This survey established an important basis from which acclimatisation societies could provide tribunals and courts with comparisons of the values that rivers under threat have to anglers. Although the evaluations provided by anglers were subjective, it was argued (and eventually accepted) that cumulatively they provided a basis for objective comparisons of the values ascribed to the various rivers under discussion.

USE OF FISH IN BIOLOGICAL CONTROL

The advent of water quality problems in some New Zealand lakes was indicated by the excessive growths of aquatic macrophytes, usually exotic species. In some lakes, macrophyte beds were eventually replaced by phytoplankton blooms as eutrophication became more serious. This was particularly a problem in Rotorua, though it was by no means the only example (e.g., Lake Tutira in Hawkes Bay). Apart from research aimed at controlling, mitigating, or avoiding such problems, attention was given to the use of fish

for biological controls of such plants. Asiatic carps were examined. Grass carp (*Ctenopharyngodon idella*) were introduced by Auckland University in 1966 and by Government in 1971; the latter imports began a substantial investigation of both the value of grass carp in controlling macrophytes and the possible impacts of these fish in New Zealand aquatic environment (Edwards & Hine 1974; Edwards & Moore 1975; Mitchell 1980, 1986; Rowe 1984). These studies culminated in a detailed environmental impact report (Rowe & Schipper 1985) and though this report recommended that the fish posed minimal danger to the environment and could perform a useful task in weed control, their use remained controversial. Opponents were unwilling to accept that alternative methods of drain clearance caused greater ecological damage; nor did some accept that the likelihood of natural reproduction in our waterways was slim. Confusion was generated by pointing to the known impacts of the quite different European carp (*Cyprinus carpio*). Ultimately, a decision was made that the use of grass carp in weed control would continue, but using only sterile triploid fish. These are now being produced and are routinely used in many parts of northern New Zealand for weed control (McCarter 1988).

Silver carp (*Hypophthalmichthys molitrix*) were also brought to New Zealand, initially by the Hawkes Bay Acclimatisation Society. The society found the work beyond its capacity and asked that Government assume responsibility; thus began a research programme on the potential impacts and value of silver carp here. Unlike grass carp, silver carp were shown to have minimal value in the restoration of the quality of lakes as trout habitats though they did improve the appearance of affected lakes (Carruthers 1986). For reasons that are difficult to understand, silver carp escaped the criticism directed at grass carp; thus silver carp were regarded as of little use and of little danger whereas grass carp were found to be valuable but were classed as an unacceptable environmental risk by some commentators.

INDIGENOUS FISH

As discussed above, discovery of New Zealand's indigenous fish fauna began with the earliest explorers and then colonists reaching this area of the world. Progress in understanding the fauna was slow, since most early emphasis was placed on exotic salmonids. William Phillipps of the Dominion Museum provided some broad summations of the fauna during the 1920s

(see references in Richardson & McDowall 1987), but knowledge of the fauna advanced slowly. Then Gerald Stokell of North Canterbury, beginning with studies of salmonids in Lake Ellesmere, quickly moved towards much more extensive taxonomic revisionary studies of the native fauna that lasted from the late 1930s until 1973. Stokell was an amateur naturalist with a limited understanding of taxonomy, but he nevertheless had a good eye for a species and he was evidently energetic in the field; he was responsible for having first named 8 of 27 fish species recognised in the fauna at present (McDowall 1990). It was on Stokell's foundation work (McDowall 1975a) that present knowledge of the fauna is based. There have since been extensive taxonomic revisions of the fauna, a few additional species have been described, and our understanding of the fauna is now more refined, ordered, and disciplined (McDowall 1962, 1970, 1975b, 1979, 1990). But it is not fundamentally different from Stokell's legacy from the 1940s through 1960s.

Although a good taxonomic insight in the fauna is essential, it is on an understanding of life histories, behaviour, and ecology of the fishes that an ability to manage species for exploitation or conservation is based. With regard to the native fishes, in this area most progress has been made in recent decades. In the early 1960s, for instance, it was not even known that bullies (*Gobiomorphus* spp.) are sexually dimorphic, with the males different in shape and coloration from females (the two sexes of one species were sometimes identified as different species). Neither was it known that five *Galaxias* species contributed to the whitebait fishery; nor had anyone reported that some bullies and the torrentfish spend their larval and juvenile lives at sea. The reproductive habits of only one species, inanga, were known in any detail—a result of some early 1900s observations: McKenzie 1904; Phillips 1919, 1924b; repeated by Hayes in the 1930s: Hefford 1931, 1932, 1933. Typical habitats occupied by species were not described, nor were their environmental tolerances and preferences; migration patterns were unknown; foods utilised were not investigated; parasites had not been examined. There was still an open field awaiting adventurous and energetic research biologists in the early-mid 1960s.

It was into this waiting opportunity that several new young fisheries biologists were deployed by the Marine Department, and in which graduate students at our universities could focus their attentions, and this is what happened. The past two decades have seen an enormous growth in the knowledge of the

native fish fauna, much of it done by Government fisheries biologists in the Ministry of Agriculture and Fisheries, with various universities making significant contributions. In the early years there was work by Hillary Jolly and Hector McMillan (both on retropinnid smelts) and somewhat later on fishes by Cedric Wood (various groups), Vivienne Benzie, Charles Mitchell and Martin Bonnett (galaxiids), Philip Cadwallader (galaxiids), Derek Staples and Chris Hatton (bullies), Ceri Hopkins (various galaxiids and bullies), Tony Eldon (mudfishes), Theo Stephens (bullies and smelts), Gordon Glova and Paul Sagar (stream ecology, and fish behaviour and feeding), and Michael Hine (fish parasites), as well as my own work on nearly all the native fauna (the very numerous papers by all these workers, up to the mid 1980s, are listed in Richardson & McDowall 1987; many of the more recent papers in McDowall 1990). Such has been the accumulation of information that detailed accounts of the fishes and their life histories are now possible (McDowall 1990), although inevitably much remains to be learned.

STUDIES OF STREAM ECOLOGY

A further field of investigation that justifies mention in this historical review is studies of stream fauna. Some of these examined faunas in their own right: Hopkins, for instance, reported on feeding and biological production in small Wairarapa streams (Hopkins 1965, 1970, 1971). However, there has been relatively little synthetic work on fish in stream ecosystems and fish inter-relationships, and this is a field of endeavour for the future. In recent years changes to the economic structure of New Zealand, with the removal of subsidies and demands that projects be economically viable, have released pressure from freshwater fisheries research that may provide the opportunity for the development of a synthetic approach to stream fish ecosystems. Work at Otago University, where a long-term research project on the fish populations of the Taieri River is underway, potentially provides additional opportunities for such studies—including much-needed investigations of the impacts of exotic and indigenous fishes (see below).

Work on the feeding ecology of native fishes, relationships between the fish and the stream benthos, and of fish feeding behaviour by Gordon Glova and Paul Sagar (Glova & Sagar 1989a, 1989b, 1991) are beginning to provide understanding of the way these ecosystems function. Fundamental to all these are

studies of the stream benthos itself, work which lies outside of this review, and which has been pursued with energy by scientists at the University of Canterbury under the leadership of Michael Winterbourn (see Winterbourn 1991: this issue).

Related to stream ecology, a few studies have examined the impacts of pesticides that were widely used in New Zealand in former times, some of them still being used. These studies may have been provoked partly by contemporary concern about the insidious and enduring effects of organochlorine insecticides that followed publication of "Silent Spring" (Carson 1962). Specifically, Hopkins et al. (1966) examined the impacts on a stream fauna of the use of DDT-impregnated fertiliser prills, aerially spread on farming pasture in a stream catchment in the Wellington district; Burnet (1972) examined the use of the herbicide paraquat to eliminate aquatic macrophytes from a small stream. In both instances significant impacts on stream fauna were reported.

INTERACTIONS OF NATIVE AND EXOTIC FISH SPECIES

As a result of early acclimatisation society activities in New Zealand (Thomson 1922), a substantial number of exotic fish species was brought to New Zealand, beginning in the 1860s, and importations continued for over 100 years (McDowall 1990). Several introduced species became abundant and widespread, most notably brown and rainbow trout and quinnat (chinook) salmon. All three species are predatory carnivores, though quinnat salmon spend most of their lives, and all of their feeding lives at large size, at sea. The early colonists and acclimatisers were concerned to see fish species useful for angling and for food in our rivers, finding few of the native species of great value. Considering the abundance and quality of fresh waters available, and coming as they did from cool temperate Great Britain, they naturally chose to bring large salmonids here.

There were early allegations that predation by the trout species was causing a decline in abundance of native lacustrine species, like koaro (Hiroa 1921; Best 1929; see McDowall 1987, 1990 for reviews of these matters). However, most of these discussions were based on circumstantial evidence and explicit documentation on impacts was lacking. This is predictable, since during the early days of the establishment of salmonids, there was little or no concern for the potential impact on the native fauna—though there was a recognition that the availability of

small fishes in New Zealand lakes and estuaries provided the salmonids with a very rich and productive food supply (Phillipps 1925; Hefford 1927). More recently studies initiated at Otago University, by Colin Townsend and his colleagues, have begun to examine questions relating to the interactions of native and exotic species; these have begun to show some explicit impacts of brown trout on common river galaxias (McIntosh 1990), as well as distinctly complementary distributions of these two species in the Taieri River (Townsend & Crowl 1991).

THE FRESHWATER FISH DATABASE

Over the past 30 or more years, especially since electric-fishing technology has been freely available to fisheries workers in New Zealand, a very large number of collecting stations has been studied. They vary from ad hoc spot sampling to detailed and repeated examinations of the fish populations of a location. It became obvious during the early 1970s that much of the information being collected was in danger of, at best, being locked away in the records of individual scientists, and at worst, being lost altogether. To prevent this loss, to enable the accumulation of an historic database on fish distributions and habitats, and to make such distributional and habitat information available to all workers in freshwater fisheries, the New Zealand Freshwater Fish Database was established. Workers in freshwater fisheries from all New Zealand institutions were invited to contribute data, and carefully structured data cards were provided for anyone interested in contributing. At the inception of the database, attempts were made to recover historic records, and the earliest of them go back to the early 1960s. During the past 10–12 years, this database has accumulated data for nearly 10 000 locations in all parts of New Zealand; it now provides a record of fish occurrences that forms the basis for our understanding of fish distributions (McDowall & Richardson 1983, 1986).

THE FUTURE FOR FRESHWATER FISHERIES RESEARCH

Where freshwater fisheries research proceeds from here is a matter of wide conjecture. The future will be profoundly affected by the institutional arrangement of scientific research in New Zealand in general, and in particular by a recent decision that freshwater fisheries research presently in the Ministry of

Agriculture and Fisheries will be transferred to the Atmosphere and Water National Research Institute. The establishment of contestable pools for the funding of research will also have significant impacts. Setting aside these bureaucratic/institutional points, future research seems likely, to some extent, to move away from recent emphases on environmental impact-driven studies to those that focus more on the processes that control the aquatic communities of New Zealand rivers and lakes. New Zealand has a particularly small freshwater fish fauna, the native and introduced species not totalling 50 species (McDowall 1990). By comparison with many lands, they are well known; yet for many only sparse details are available on life cycles, reproductive, behaviour, habitat requirements, environmental tolerances, and so on. Even less is known about community structures, which factors control populations, growth rates, and so on.

However, present requirements that research agencies should earn a significant portion of their revenue will force research groups to undertake research commissioned and funded by agencies with specific interests; these are likely to result in some staff, or some staff time, focusing back on environmental issues. In particular, likely increases in forestry activities are bound to have environmental impacts that need investigation, prediction, and mitigation, if the effects of forestry on in-stream values are to be dealt with responsibly. Wood processing industries are also likely to expand, with their recognised but unquantified concomitant impacts on aquatic values. Eventually, there will need to be a return to construction of hydro dams and possibly of thermal power stations. Thus, although pressures for environmental impact studies seem less urgent now than during the early 1980s, a return to such studies seems inevitable.

We need rapidly to increase our understanding of: the factors that affect the distribution and abundance of native fish species; what controls the fish communities that occupy various habitat types; the way energy flows through the ecosystem; the role of different types of vegetative cover in affecting the fish populations in streams draining and passing through that vegetation.

Obtaining this information will put decision-makers in New Zealand in a much stronger position:

- to understand the impacts of pastoral farming and horticulture on stream fauna;
- to evaluate the impacts of monoculture of *Pinus* or other exotic trees, and the consequences of periodic milling of such forests;
- to predict what will happen to fish populations when rivers are impounded or abstracted;
- to improve understanding of the impacts of existing exotic fish species and predict the impacts of others proposed for introduction or translocation;
- to better manage fish populations under exploitation by recreational and commercial fisheries;
- to ensure the reliable conservation of all indigenous species, especially those that are at risk; and
- to better fulfil Government's responsibilities to Maori, with regard to freshwater fisheries, under the Treaty of Waitangi.

All of these are highly desirable goals and, at present, there seems the opportunity to make significant progress towards achieving them. We should ensure that this opportunity is exploited to the full.

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